Mathematics and Statistics Curriculum

Algebra, graphs, percentages, statistics, probability, and mathematical reasoning.

Reference material for lesson planning and course-content development.

Contents

1. Teaching mathematics through worked reasoning

Mathematics lessons should make the reasoning visible, not just the final answer. A worked example is useful when each step has a reason. Learners should know why an operation is allowed, what quantity is changing, and how the result can be checked. Avoid examples where the teacher performs six symbolic steps without explanation and then asks learners to copy the pattern.

Use small numbers when introducing a new structure. Complexity can come later. The first goal is to help learners see the relationship, not to test arithmetic endurance.

- Ask for estimates before exact calculation when it helps check reasonableness.
- Use diagrams, number lines, tables, or graphs when they expose the structure.
- Include at least one common wrong path and explain why it fails.

Checking is part of mathematics. Substitution can check an equation solution. Recalculating with another method can check a percentage. Reading a graph can check whether a slope sign makes sense.

2. Linear equations

An equation states that two expressions have equal value. Solving means finding the value or values that make the statement true. The key idea is preserving equality. If an operation is applied to one side, the corresponding operation must be applied to the other side.

One-step equations

Begin with equations such as $x + 7 = 12$ or $4x = 20$. Learners should name the inverse operation before doing it. The goal is not to memorize 'move it to the other side'. That phrase can hide the actual operation.

Two-step equations

Use examples such as $3x + 5 = 20$. The learner subtracts 5 from both sides, then divides both sides by 3. Have learners check by substituting the solution into the original equation.

Common mistakes

- Changing a sign without applying an operation to both sides.
- Dividing only one term instead of the whole side.
- Stopping after finding a value without checking it.
- Combining unlike terms.

Word problems should be short enough that the equation is the main challenge. Ask learners to define what the variable represents before writing the equation.

3. Slope and linear graphs

Slope describes how much y changes when x changes. Learners should see slope both as a ratio and as a visual feature of a line. Use two points, calculate rise over run, then connect the sign of the result to the direction of the graph.

Interpretation

A slope of 3 in a savings graph might mean savings increase by 3 units for each unit of time. A slope of -2 in a water-tank graph might mean the level drops by 2 units per minute. Units should be included in interpretation questions.

Y-intercept

The y-intercept is the value of y when x is zero. In $y = mx + b$, b is the y-intercept. It may represent a starting amount, initial fee, or baseline measurement depending on the context.

Common mistakes

- Reversing rise and run.
- Using points in inconsistent order in numerator and denominator.
- Assuming a negative slope means one of the coordinates must be negative.
- Confusing the y-intercept with any point where the line crosses a grid line.

Give learners both graphs and tables. They should be able to identify a constant rate of change from either representation.

4. Quadratic expressions and graphs

Quadratic expressions contain a squared variable term. Introductory work should connect algebraic forms to graph behavior without turning the lesson into a complete algebra course.

Expanding and factoring

Use patterns such as $(x + 3)(x + 2)$ and $x^2 + 5x + 6$. Learners should connect factors to values that make a product zero. Factoring should be presented as one useful form, not as a trick disconnected from the original expression.

Zeros and graph shape

The zeros of a quadratic are x-values where the graph crosses or touches the x-axis. A parabola can have two real zeros, one repeated real zero, or no real zeros. The opening direction depends on the sign of the squared-term coefficient.

Discriminant awareness

At an intermediate level, the discriminant can be used to predict the number of real solutions. The lesson does not need a proof, but learners should know what positive, zero, and negative values imply.

- Keep early coefficients small enough to factor cleanly.
- Ask learners to compare factored and expanded forms.
- Include a graph interpretation item, not only symbolic manipulation.

5. Ratios, rates, and percentages

Ratios compare quantities. Rates compare quantities with different units. Percentages express a part per hundred. These ideas appear across business, science, data analysis, and everyday decisions, so examples should include more than shopping discounts.

Percentage change

Percentage change compares the difference to the original value. Learners often divide by the new value instead. Use a table showing original, new, difference, and percentage change.

Percentage points

A move from 40 percent to 50 percent is an increase of 10 percentage points, not a 10 percent increase. The relative increase is 25 percent. This distinction is useful when reading reports.

Rates

Examples include kilometers per hour, requests per second, cost per item, and students per teacher. Units help learners see what is being divided by what.

- Include a reverse-percentage problem after basic examples.
- Ask whether a result is plausible before accepting the arithmetic.

6. Mean, median, mode, range, and outliers

Measures of center summarize a dataset in different ways. The mean uses every value, the median depends on order, and the mode identifies the most frequent value. A dataset may have no mode or more than one mode.

Outliers

A large outlier can pull the mean away from the typical cluster while leaving the median relatively stable. This makes a good comparison exercise. Give learners a small dataset, calculate both measures, add one extreme value, then calculate again.

Dataset	Mean	Median	Comment
4, 5, 5, 6, 7	5.4	5	Values are close together
4, 5, 5, 6, 27	9.4	5	Mean is pulled upward

Range

Range is the difference between the maximum and minimum values. It is simple but depends heavily on extremes. It can be useful as a first measure of spread before standard deviation is introduced.

Learners should explain why they chose a particular summary measure for a context rather than treating one measure as always best.

7. Reading charts and tables

Charts are not only pictures of data. They are arguments about what information deserves attention. Learners should be able to read values, compare categories, identify trends, and notice design choices that can exaggerate or hide differences.

Bar charts

Bar charts compare categories. The vertical scale matters. A truncated axis can make a small difference look dramatic. Learners should read the actual values rather than relying only on visual height.

Line charts

Line charts are useful for change over an ordered dimension such as time. Ask learners whether the trend is steady, accelerating, irregular, or flat. Avoid assuming every line chart proves causation.

Tables

Tables can show exact values more clearly than charts. Learners should practice locating a row, applying a filter condition, and comparing two columns.

- Include one misleading axis example.
- Include one question where the chart does not provide enough information for the requested conclusion.
- Ask learners to write one sentence that is supported and one sentence that is not supported by the chart.

8. Introductory probability

Probability describes uncertainty. Begin with outcomes learners can list, such as coins, dice, colored tokens, or simple event counts. The language of event, outcome, impossible, certain, and equally likely should be clear before formulas become central.

Simple probability

For equally likely outcomes, probability can be calculated as favorable outcomes divided by total outcomes. This assumption should be stated. Not every real-world event has equally likely outcomes.

Independent events

At an introductory level, show that one coin toss does not change the physical probability of the next fair toss. This helps address the common belief that a run of heads makes tails 'due'.

Using data

Experimental probability comes from observed frequency. It may differ from theoretical probability in a small sample. Larger samples often stabilize, but they do not guarantee an exact theoretical proportion.

- Ask learners to compare theoretical and experimental results.
- Avoid gambling contexts if they distract from the concept.

9. Functions and input-output thinking

A function associates each allowed input with one output. This idea connects tables, formulas, graphs, and programming. Early examples can use machines that transform a number, but the lesson should move quickly to normal notation and real relationships.

Tables

An input-output table helps learners see a pattern before writing a formula. Ask what changes consistently and whether every input has one output.

Graphs

Graphs show many input-output pairs at once. The vertical-line test can be introduced visually when learners are ready to distinguish functions from other relations.

Formulas

For $f(x) = 2x + 3$, learners should calculate values, interpret the operation sequence, and connect the result to a point on the graph.

Avoid turning function notation into a vocabulary exercise. The learner should use it to answer questions and compare representations.

10. Designing mathematics practice

Practice should not be twenty copies of the same calculation. A useful set mixes direct application, error analysis, interpretation, and one transfer problem. Learners need enough repetition to become fluent, but repetition should not hide whether they understand the underlying idea.

A balanced set

- Two straightforward problems to confirm the basic method.
- One problem where the learner predicts or estimates first.
- One worked solution containing an error to diagnose.
- One context problem that requires choosing the method.
- One explanation question.

Answers should include reasoning where a common misconception is likely. For open-ended questions, a marking guide can list the important features rather than one exact sentence.